

Complete Technical Modification & Implementation Report (Role Request & Admin Approval Workflow)

1. Executive Summary & Objective (■■■■■■■■ ■■ ■■■■■■)

Component / Requirement	Implementation Details & Status
1. New Employees Stored	Vilash Deshmukh (MPO107 - TL) & Rajesh Kumar (MPO112 - Senior Software Engineer) registered in EmployeeHubDb & synced to Knome.
2. Zero-Role Detection	When new user logs in without an approved role, system marks roleStatus as Pending .
3. Role Request Generation	Automatic insertion into RoleRequests queue table upon initial login.
4. EmployeeHub Pending UI	Dedicated RolePending.tsx screen with real-time status polling and unlock transition.
5. Knome System Admin Approval	New Role Requests tab in Knome AdminConsole.jsx for 1-click role assignment.
6. Dual-Database Synchronization	Approval immediately updates EmployeeRoles (EmployeeHubDb) & UserRoles (Knome).
7. Identity Collision Fix	Fixed fallback logic that previously redirected MPO107 to MPO101 (Loveneesh Sharma).

- **Default Roster Missing MPO107:** Frontend ███ DEFAULT_EMPLOYEES ███ INITIAL_USERS ██████████ MPO101 ███ MPO106 ███ ███ ██████████ MPO107 ██████████ emps.find(...) || emps[0] ██████████ emps[0] (Loveneesh Sharma) ██████████
- **Immediate Knome SSO Redirect:** Login ██████████ Knome (Port 5173) ██████████, ██████████ localStorage ██████████ MPO101 ██████████
- **(Resolution):** MPO107 ███ DEFAULT_EMPLOYEES, INITIAL_USERS ███ API Fallback ██████████ Login.tsx ██████████ ██████████ ██████████ Knome ██████████ EmployeeHub ███ Role Pending Screen ██████████

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```
-- RoleRequests Table Creation in EmployeeHubDb & Knome
CREATE TABLE [dbo].[RoleRequests] (
[RequestId] INT IDENTITY(1,1) NOT NULL PRIMARY KEY,
[EmployeeId] VARCHAR(50) NOT NULL,
[FullName] NVARCHAR(150) NOT NULL,
[Email] VARCHAR(150) NOT NULL,
[DepartmentId] INT NOT NULL,
[DepartmentName] NVARCHAR(100) NULL,
[Designation] NVARCHAR(100) NOT NULL,
[RequestedRoleCode] VARCHAR(30) NOT NULL DEFAULT 'EMP',
[Status] VARCHAR(30) NOT NULL DEFAULT 'Pending', -- 'Pending', 'Approved', 'Rejected'
[AssignedRoleName] NVARCHAR(50) NULL,
[AssignedBy] VARCHAR(50) NULL,
[CreatedAt] DATETIME2(7) NOT NULL DEFAULT GETUTCDATE(),
[ProcessedAt] DATETIME2(7) NULL
);
```

Layer 5 — Backend Services (EmployeeHub.API & Knome.API)

EmployeeHub.API:

- EmployeeHub.Domain/Entities/RoleRequest.cs: ■■■■ ■■■■■■ ■■■■ ■■■■■■ ■■■■■■ ■■■■■■ ■■■■■■
- EmployeeHubDbContext.cs: DbSet<RoleRequest> RoleRequests ■■■■■■■■ ■■■■■■
- AuthService.cs: ■■■■■■ ■■ ■■■■■■-■■■ ■■■■ ■■ ■■■■ RoleRequests ■■■■■■■■ ■■ RoleStatus='Pending' ■■■■■■■■
- EmployeeController.cs: GET /api/employees/role-requests, POST /api/employees/role-requests/{id}/approve ■■■■■■■■■■ ■■■■■■■■

Knome.API:

- [illegible]

Layer 1 — Presentation Layer (EmployeeHub.Web & Knome UI)

EmployeeHub.Web (Port 5001):

- [illegible]

Knome UI (Port 5173):

- `pages/AdminConsole.jsx`: **Role Requests (N)** (Initial Users) + Approve & Assign
- `utils/apiService.js`: `roleRequestsApi`, `adminApi.getRoleRequests`, `adminApi.approveRoleRequest`
- `components/contexts/UserContext.jsx`: `MPO107` INITIAL USERS

4. End-to-End Verification & Testing Guide (■■■■■■■■■■ ■■■■■■■■■■)

Step #	Action / Operation	Expected Screen / Result
Step 1	Open <code>http://localhost:5001</code> and enter MPO107 / Password@123 .	Opens Role Assignment Pending Approval screen showing Vilash Deshmukh, TL, Development.
Step 2	Open <code>http://localhost:5173</code> , login as MPO101 (Loveneesh Sharma).	Access Knome Admin Console -> Navigate to Role Requests tab.
Step 3	Select Role (e.g. System Admin / Employee) & click Approve & Assign .	Request marks as Approved; updates EmployeeHubDb & Knome databases.
Step 4	Switch back to EmployeeHub (Port 5001) or click Check Status .	Displays celebratory Role Successfully Assigned! and unlocks Dashboard.

5. Summary of Modified & Created Files

Action	File Path	Description
[NEW]	EmployeeHub/sql/Add_RoleRequests_Table.sql	SQL table creation and initial seed data for MPO107.
[NEW]	EmployeeHub/sql/Add_Employee_Vilash_Deshmukh.sql	Master SQL insert for Vilash Deshmukh.
[NEW]	EmployeeHub/src/EmployeeHub.Domain/Entities/RoleRequest.cs	C# Entity model for RoleRequest.
[NEW]	EmployeeHub/src/EmployeeHub.Web/src/pages/RolePending.tsx	Role Pending status page with live polling.
[MOD]	EmployeeHub/src/EmployeeHub.Application/Services/AuthService.cs	Zero-role auto-request detection on login.
[MOD]	EmployeeHub/src/EmployeeHub.API/Controllers/EmployeeController.cs	Role requests and status check endpoints.
[MOD]	EmployeeHub/src/EmployeeHub.Web/src/App.tsx	RolePending view routing.
[MOD]	EmployeeHub/src/EmployeeHub.Web/src/services/api.ts	Added MPO107 & role request API clients.
[MOD]	Knome/Backend/Knome.API/Services/UserService.cs	Dual-database role synchronization logic.
[MOD]	Knome/knomeUI/frontend/src/pages/AdminConsole.jsx	Role Requests tab & approval UI.
[MOD]	Knome/knomeUI/frontend/src/utils/apiService.js	RoleRequests API service layer.
[MOD]	Knome/knomeUI/frontend/src/components/contexts/UserContext.jsx	Added MPO107 to initial roster & session restore.